

Andrea Reguitti

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Research interests

- Time-domain stellar astrophysics; optical transients; supernovae: interacting, core-collapse, thermonuclear; Gap Transients: supernova impostors, stellar mergers, electron-capture supernovae candidates.
- Electromagnetic domains: Near-Ultraviolet (NUV), Optical, Near-Infrared (NIR), Mid-Infrared (MIR).

Education

- 04/2019 – **Ph.D. in Astrophysics**, *Universidad Andres Bello (UNAB)*, Santiago, Chile, Supervisor: Dr. G. Pignata
06/2023 (UNAB), Co-supervisor: Dr. A. Pastorello (INAF-OAPd).

Thesis title: Exploring the final stages of massive stars through Type II_n Supernova precursors and LBV variability
In my PhD thesis I searched for precursor activity years before the explosion of a sample of 27 nearby Type II_n SNe. Then, I reconstructed decades-long light curves of a sample of 12 eruptive Luminous Blue Variables in the Local Group, looking for their eruptions. Finally, I compared the light curves of the precursors to those of the LBVs, to see if they are similar and therefore if LBVs are indeed good candidates as progenitors of SNe II_n. While the characteristics of the precursors and the LBV eruptions are different, I still found a few similarities, making the connection possible.

- 10/2016 – **M.Sc. in Astronomy**, *Padova University*, Padova, Italy, Advisors: Dr. S. Ciroi and Dr. A. Pastorello,
09/2018 *final remark - 110/110 cum Laude*.

Thesis title: The peculiar SN 2013gc: evidences of an eruptive phase before the explosion
In this thesis I analysed the interactive Type II_n SN 2013gc, its light curves and spectra. By looking at archival images taken a few years before the explosion, I found evidences of an eruptive phases from the progenitor, during which it expelled circumstellar material, with which the SN ejecta interacted after the explosion.

- 10/2013 – **B.Sc. in Astronomy**, *Padova University*, Padova, Italy, Advisors: Dr. S. Ciroi and Dr. M. Berton,
10/2016 *final remark - 101/110*.

Thesis title: Determinazione della massa del buco nero in un campione di FSRQ a redshift intermedio
In this thesis I studied a small sample of 9 flat-spectrum radio quasars at intermediate redshift, analysing their spectra from the SDSS survey, and from them I calculated the masses of the supermassive black holes at their centres.

Employment history

- 07/2025 – present **Postdoc**, *Assegno di ricerca "Attività scientifica e di supporto con SOXS, strumento del telescopio ESO-NTT"*, INAF-PADOVA OBSERVATORY, Padova, Italy.

Advisor: Dr. R. Claudi

- 07/2023 – **Postdoc**, *Assegno di ricerca "Attività di supporto alle attività della collaborazione GRAWITA durante O4; interpretazione ed analisi dati multifrequenza di controparti di sorgenti GW e di altre sorgenti transienti"*, INAF-BRERA OBSERVATORY, Merate (LC), Italy.

Advisor: Dr. P. D'Avanzo

- 08/2018 – **Formative stage**, INAF-PADOVA OBSERVATORY, Padova, Italy.

11/2018 Advisor: Dr. A. Pastorello

Languages knowledge

Native language **Italian**.

Second language **English**, Reading ability: Excellent; Writing ability: Very good; Speaking ability: Very good.

Third language **Spanish**, Reading ability: Basic; Speaking ability: Intermediate.

Publications summary

- Total: 86 refereed publications (8 as first-author), 13 submitted; >250 non-refereed publications; ~1750 citations. H-index: 24.
- ADS public library URL: <https://ui.adsabs.harvard.edu/public-libraries/in38wuqjTi6q0-0ZzVnzQA>
- Google Scholar public URL: https://scholar.google.com/citations?user=Jw_zBMoAAAAJ&hl=it&oi=sra

Scientific Collaborations

- 02/2026 – now **Member of the Network INAF for Cosmic Explosions in the Time-domain and Multi-messenger Era (NICETIME).**
- 07/2024 – now **Member of the Coordinamento Italiano Burst Ottici (CIBO) Collaboration.**
- 06/2024 – now **Junior Member of the International Astronomical Union (IAU).**
- 07/2023 – present **Member of the Electromagnetic counterparts of gravitational wave sources at the Very Large Telescope (ENGRAVE) Collaboration.**
- 7/2023 – 2/2026 **Member of the GRAVitational Wave Inaf TeAm (GRAWITA) Collaboration, search and follow-up of electromagnetic counterparts of gravitational waves events.**
- 06/2022 – now **Member of the LSST Transients and Variable Stars (TVS) Science Collaboration.**
- 01/2020 – present **Associated to the Istituto Nazionale di Astrofisica (INAF) - Osservatorio Astronomico di Padova (OAPd).**
- 6/2019 – 4/2023 **Affiliated to the Chilean Millennium Astrophysical Institute (MAS).**
- 1/2018 – now **Member of the Public ESO Spectroscopic Survey for Transient Objects (PESSTO), to study optical transients with the NTT telescope, member of the interacting supernovae group, member of the Target Alert Team for 11 subruns.**
- 12/2017 – present **Member of the NOT Unbiased Transient Survey (NUTS), to study optical transients with the NOT telescope, duties of sending triggers and reducing data.**

Honors and awards

- 2018 **Mille e una lode**, award for the student with the highest average exams score in the Academic year 2016-2017, PADOVA UNIVERSITY, 1,000 EUR.

Scientific presentations (last 3 years)

- 30/09/2025 **SOXS Data Sharing**, "SOXS Operation teams meeting", Brera Astronomical Observatory, Milan, Italy, (contributed talk).
- 20/06/2025 **Using space telescopes to study transients**, "ASIAGO OAPd Day", Asiago Astrophysical Observatory, Italy, (presentation talk).
- 02/04/2025 **The fate of the progenitors of Luminous Red Novae**, *An Extraordinary Journey Into The Transient Sky: from restless progenitor stars to explosive multi-messenger signals*, conference, (contributed talk). consultable here: https://www.memsait.it/videomemorie/volume-2-2025/VIDEOMEM_2_2025.28.mp4
- 11/03/2025 **Studying Intermediate Luminosity Red Transients and their progenitors with Space Telescopes**, "Transients From Space" workshop, Baltimore, Maryland, USA, (spark talk).
- 26/02/2024 **Scientific contributions of the telescopes of the Asiago Observatory**, *6th Chianti Topics – International Focus Workshop*, (contributed talk). consultable here: https://www.memsait.it/videomemorie/volume-1-2024/VIDEOMEM_1.2024.03.mp4
- 23/01/2024 **Precursors of Type IIn Supernovae in the era of LSST**, *What Was That? – Planning ESO Follow Up for Transients, Variables, and Solar System Objects in the Era of LSST*, (poster and lightning talk).
- 06/11/2023 **The double peaked light curve of SN 2016aiy**, *online Supervirtual conference 2023*, (poster).
- 21/09/2023 **The double-peaked light curve of the Type IIn SN 2016aiy**, *ePESSTO+ Meeting*, Warsaw Astronomical Observatory, Warsaw, Poland, (contributed talk).
- 07/09/2023 **A bunch of Type Ibn SNe followed with NUTS**, *3rd NUTS2 Meeting*, Padova University, Padova, Italy, (contributed talk).

15/06/2023 **L'universo transiente al Copernico**, *Copernico @50 workshop*, Asiago, Italy, (contributed talk).
consultable here: https://www.memsait.it/videomemorie/volume-3-2023/VIDEOMEM_3.2023.31.mp4

Participation at conferences, workshops, meetings (last 3 years)

- 10/02/2026 – **Multimessenger astronomy in the Einstein Telescope Era**, *GRAWITA collaboration meeting*,
12/02/2026 Palazzo della Salute, Padova, Italy.
- 29-30/09/2025 **SOXS Operation teams meeting**, *2-days workshop*, Brera Astronomical Observatory, Milan, Italy.
01/07/2025 **SOXS DAY**, *1-day meeting*, ESO HQ, Garching bei Munich, Germany.
- 23/06/2025 – **2025 European Astronomical Society (EAS) Annual Meeting**, *conference*, University College
27/06/2025 Cork, Cork, Ireland.
- 20/06/2025 **ASIAGO OAPd Day**, *1-day workshop*, Asiago Astrophysical Observatory, Asiago Italy.
- 01/04/2025 – **An Extraordinary Journey Into The Transient Sky: from restless progenitor stars to explosive
04/04/2025 multi-messenger signals**, *conference*, Palazzo della Salute, Padova, Italy.
- 24-28/3/2025 **Celebrating 20 years of Swift Discoveries**, *Conference*, Firenze Fiera, Florence, Italy.
- 11-13/3/2025 **Transients From Space**, *workshop*, STScI, Baltimore, Maryland, USA.
- 26-27/9/2024 **L'Eredità scientifica di Giuseppe "Bepi" Colombo**, *convegno*, Palazzo Bo, Padova, Italy.
- 01-05/07/2024 **2024 EAS Annual Meeting**, *conference*, Padova Congress Hall, Padova, Italy.
- 25-27/6/2024 **SOXS Consortium Science Meeting**, INAF-Osservatorio Astronomico di Capodimonte, Naples, Italy.
- 26/02/2024 – **Use of small telescopes in the giant era II**, *Chianti Topics VI*, *workshop*, Complesso di Sant'Apollonia,
29/02/2024 Florence, Italy.
- 22/01/2024 – **What Was That? – Planning ESO Follow Up for Transients, Variables, and Solar System
26/01/2024 Objects in the Era of LSST**, *workshop*, ESO Headquarters, Garching bei München, Germany.
- 6-10/11/2023 **online Supervirtual conference 2023**, virtual conference on supernovae and related transients.
- 23/10/2023 – **Anisotropies in core-collapse supernova explosions 2**, *conference*, Istituto Euro-Mediterraneo di
26/10/2023 Scienza e Tecnologia, Palermo, Italy.
- 21-22/9/2023 **ePESSTO+ Meeting**, Warsaw Astronomical Observatory, Warsaw, Poland.
- 7-8/9/2023 **3rd NUTS2 Meeting**, Osservatorio Astronomico di Padova, Padova, Italy.
- 14-16/6/2023 **Copernico @50**, *workshop*, organised by L. Tomasella, Asiago, Italy.

Recent observing time awarded (last 2 years)

- 04/2026 – **Telescopio Nazionale Galileo (TNG) 3.58m + DOLORES (30 hours) and Rapid Eye Mount
09/2027 (REM) 0.6m + ROSS+REMIR (30 hours)**, *3 semesters Large Program*, "Interacting transients:
an opportunity to investigate the mysteries of stellar evolution", La Palma Observatory.
PI: G. Valerin
- 02/2026 – **Copernico 1.82m + AFOSC (3 nights/month), Schmidt 0.67m + Moravian (40 hours/month),
01/2028 2 years Large Program**, "Extended monitoring of peculiar stellar transients with Asiago telescopes",
Asiago Astrophysical Observatory.
PI: A. Reguitti
- 04/2026 – **ESO Very Large Telescope (VLT) 8.2m UT3 + X-Shooter (5.75 hours)**, *Period 117*, "Early
09/2026 Warning Signs: Observations of Progenitor Activity Prior to Core-Collapse Supernovae", Cerro Paranal
Observatory.
PI: S.J. Brennan
- 04/2026 – **ESO Very Large Telescope (VLT) 8.2m UT3 + X-Shooter (16 hours) + FORS2 (8 hours) +
09/2026 HAWK-I (8 hours)**, *Period 117*, "Compact object mergers and heavy element enrichment: Kilonovae
from the VLT", Cerro Paranal Observatory.
PI: A. Leván
- 10/2025 – **NOT 2.56m (2 nights of SoftToO) + ALFOSC**, *Period 72*, "PRE-Supernova Targeted Observations
03/2026 (PRESTO) during the Rubin Era", La Palma Observatory.
PI: M. Stritzinger

- 07/2025 – **NOT 2.56m (1.5 hours), LT 2m + IO:O (8 hours) and LIRIC (7.5 hours)**, 2025B+2026A,
09/2026 "Determining the progenitor types of interacting supernovae", La Palma Observatory.
PI: **A. Reguitti**
- 10/2024 – **REM 0.6m + ROSS2/REMIR (20+20 hours)**, AOT50, "REM contribution to the worldwide search
03/2025 for an electromagnetic counterpart of a gravitational wave trigger", La Silla Observatory.
PI: S. Campana
- 09/2024 – **GTC 10.4m + HIPERCAM (1.5 nights)**, 2024B, "Identifying Progenitors of Core-Collapse Super-
02/2025 novae with HiPERCAM", La Palma Observatory.
PI: N. Elias-Rosa
- 07/2024 – **CAHA 3.5m + OMEGA2000 (12 hours), NOT 2.56m + ALFOSC (2 nights), LT 2m + IO:O
01/2025 (10 hours)**, 2024B, "Determining the progenitor types of interacting supernovae", La Palma and Calar
Alto Observatories.
PI: **A. Reguitti**
- 09/2024 – **LBT 2×8.2m + LUCI/MODS/LBC (24 hours)**, 2024B, "Spectroscopic and photometric ToO
07/2025 follow-up of gravitational wave events", Mount Graham Observatory.
PI: A. Rossi
- 10/2021 – present **Nordic Optical Telescope (NOT) 2.56m + ALFOSC and NOTCAM (65 hours/semester of
soft ToO + 12 hours/semester of hard ToO with ALFOSC, 8 half-nights with NOTCAM)**,
"The local transient Universe", La Palma Observatory.
PI: E. Kankare, T. Reynolds, M. Stritzinger
- 04/2019 – present **ESO New Technology Telescope (NTT) 3.58m + EFOSC2 and SOFI (90 nights/semester)**,
Large Program, "ePESSTO+: the (advanced) extended Public ESO Spectroscopic Survey of Transient
Objects", La Silla Observatory.
PI: C. Inserra

On-site observational experiences

- 23/07/2024 **Copernico 1.82m + Echelle high-resolution spectrograph (1 night)**, *Asiago Astrophysical
Observatory, Italy*, variable stars.
- 01/2020 – present **Copernico 1.82m + AFOSC and Schmidt 0.67m + Moravian (32 nights and counting)**,
Asiago Astrophysical Observatory, Italy, imaging and low-resolution spectroscopic observation of Gap
Transients and interacting supernovae.
- 12/2017 – **Copernico 1.82m + AFOSC and Schmidt 0.67m + SBIG (9 nights)**, *Asiago Astrophysical
04/2018 Observatory, Italy*, imaging and spectroscopic observation of supernovae.
- 06/2016 **Galileo 1.22m + B&C (4 nights)**, *Asiago Astrophysical Observatory, Italy*, for practice in the use of
a professional telescope, spectroscopic observation of quasars.

Technical and Computer skills

Processing pipelines of optical and NIR spectroscopic and photometric data, taken with Asiago
telescopes (AFOSC, Moravian), NOT (ALFOSC), NTT (EFOSC2, SOFI, SOXS), GTC (OSIRIS),
EsoReflex (X-Shooter, FORS2).

Data reduction: *Swift*, *GALEX* (NUV), *HST* (optical), *WISE*, *Spitzer*, *JWST*, *SPHEREx* (MIR)
space telescopes.

Operating System Windows, Ubuntu, MacOS

Codes language Python, LaTeX

Softwares SAO DS9, Aladin, IRAF, PyRAF, DUSTY

Professional service

- 05/26-now **Project Lead**, *ILOT working group*, ePESSTO+ collaboration.
- 05/26-06/25 **TAC member**, *LBT@INAF*, Period September 2026 - July 2027 (2026B).
- 12/2025 **Contributor**, *Dynamical binary interactions in the 2040s*, ESO White Paper for "Expanding Horizons",
PI: N. Blagorodnova, O. Pejcha, T. Kamiński.

- 20/06/2025 **LOC Member**, *ASIAGO OAPd Day*, workshop, Asiago, Italy.
- 1-4/04/2025 **LOC Member**, *An extraordinary journey into the Transient Sky: from restless progenitor stars to explosive multi-messenger signals*, conference, Padova, Italy.
- 24-28/3/2025 **LOC Member**, *Celebrating 20 years of Swift Discoveries*, conference, Florence, Italy.
- 1-5/07/2024 **Volunteer**, *2024 EAS Annual Meeting*, conference, Padova, Italy.
- 06/2023 **LOC Member**, *Copernico @50*, workshop, Asiago, Italy.
- 10/2022-present **Journal referee**, *The Astrophysical Journal*, *Monthly Notices of the Royal Astronomical Society*, *Astronomy & Astrophysics*, *Open Journal of Astrophysics*.
- 12/22-12/25 **TAC member**, *TNG/REM*, Semesters AOT47 (2023A), AOT50 (2024B), and AOT53 (2026A).

Teaching experience

- 30/07/2025- **Training** of 4 Physics MSc students from Trieste University on how to conduct an observing night
01/08/2025 with the Galileo 1.22m telescope + B&C spectrograph, Asiago Astrophysical Observatory, Italy
- 08/04/2025- **Thesis co-supervisor** of MSc in Astrophysics & Cosmology student Luca Filotto (Padova University,
10/12/2025 Italy), entitled "A detailed analysis of the Stripped-Envelope Core-Collapse Type Ib/c SN 1996aq".
- 25/07/2024- **Training** of 7 Physics MSc students from Trieste University on how to conduct an observing night
26/07/2024 with the Copernico 1.82m telescope + Echelle spectrograph, Asiago Astrophysical Observatory, Italy
- 21/07/2024- **Training** of MSc student Elisa Bigongiari (UniPi) on how to use *daophot* to measure the magnitude
22/07/2024 of a stellar source, Asiago Astrophysical Observatory, Italy
- 04/02/2024- **Tutor** of the 2023 NEON Observing School, Asiago Astrophysical Observatory, Italy
16/02/2024 https://www2.mpia-hd.mpg.de/~korhonen/GRP_training/NEON2023/
- 21/07/2023- **Training** of PhD students Beth Fitzpatrick and Robert Byrne (UCD) on how to conduct an observing
24/07/2023 run at a professional telescope, Asiago Astrophysical Observatory, Italy

Outreach Activity (last 3 years)

- 12/2020 – **ASYAGO Schools**, for high-school students, participation as a 'tutor', both online and in-person.
07/2026 Asiago Astrophysical Observatory, Italy
- 16/05/2026 **Supernovae: la fine esplosiva delle stelle**, educational colloquium, Scuola Superiore (High School)
P. Levi, Montebelluna (TV), Italy.
- 26/09/2025 **European Researchers Night**, *Supernovae: la fine esplosiva delle stelle*, organised by S. Zaggia,
INAF-Osservatorio Astronomico di Padova, Italy.
- 27/01/2025 **Supernovae: la morte violenta delle stelle**, educational colloquium, Scuola Superiore G. Perlasca,
Idro (BS), Italy.
- 02/12/2024 **Supernovae: la morte violenta delle stelle**, educational colloquium, Scuola Superiore G. Perlasca,
Vobarno (BS), Italy.
- 30/11/2024 **Supernovae: la morte violenta delle stelle**, public colloquium, Agnosine (BS) municipality, Italy.
- 06/11/2024 **Supernovae: la morte violenta delle stelle**, lecture within an Astronomy course for amateurs, Schio
(VI), Italy.
- 19/04/2024 Activity of divulgation ("divulgazione") to a children school, Asiago Astrophysical Observatory, Italy
- 29/01/2024 **I buchi neri**, educational colloquium, Scuola Superiore G. Perlasca, Idro (BS), Italy.
- 11/11/2023 **I buchi neri**, public colloquium, Agnosine (BS) municipality, Italy.
- 04/11/2023 **Il Telescopio Spaziale James Webb**, public colloquium, Lanificio Conte, Schio (VI), Italy.
- 14/10/2023 **Il Sistema Solare**, public colloquium, Agnosine (BS) municipality, Italy.
- 29/09/2023 **European Researchers Night**, oral presentation of the Asiago Schmidt telescope, organised by
S. Zaggia, INAF-Osservatorio Astronomico di Padova, Italy.

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